The Architecture of Errors: Logarithmic Mode Discovery and Polylogarithmic Intervention Budgets for Long-Context LLM Reliability

Anonymous Author(s)
Affiliation
Address
email

Abstract

Prior work on length-decay benchmarks [Anonymous, 2025] argued that LLM reliability does not decay exponentially with output length because errors concentrate at sparse "key tokens" rather than spreading uniformly. This paper extends that framework along an orthogonal axis: LLM errors are not only sparse, they are *repetitive*. The same handful of failure patterns recur across models, datasets, and domains. Inside the small set of key tokens, a latent fraction β produce *hard* failures; those hard failures cluster into a finite catalogue of recurring modes whose count grows slowly with the number of observed failures. Under an empirical postulate of logarithmic mode discovery—calibrated against endpoint counts from three published taxonomies (ErrorAtlas, MWPES, HumanEval-error-categorisation), with a direct subsample-discovery curve still outstanding—the intervention budget required to bound *per-hard-token* residual error scales polylogarithmically in sequence length. Sequence-level reliability targets are strictly tighter and approach full-catalogue coverage as the number of hard decisions grows—an asymmetry we surface explicitly rather than burying. Twelve intervention categories from prior work, plus 28 capability-elimination citations across six axes stratified into Patterns A/B/C (Appendix B), are consistent with a small library on the order of tens of interventions covering the head of the failure distribution in the per-hard-token regime. We analyze the polylog conclusion symbolically under several cluster-count laws (logarithmic, Heaps, saturating; Appendix A) and find it robust across the family, though the specific doubly-logarithmic rate is the optimistic special case. Prominent steep-decay counter-evidence [Dziri et al., 2023, Kuratov et al., 2024, Kwa et al., 2025], re-audited in Appendix C, decays over variables other than raw token length—compositional graph size, fact count, log-time horizon. This relocates rather than dissolves the practical worry: the framework identifies which interventions help and which do not, but does not claim the regimes where k_{hard} grows fast are thereby made easy.

1 Introduction

The standard worry about long-context autoregressive generation is exponential. If every token has independent error probability e, then after n tokens the chance of a fully correct output is $(1-e)^n$, which collapses to zero for any nontrivial e and large enough n [LeCun, 2023, Dziri et al., 2023]. Earlier work in this series [Anonymous, 2025] argued that this worry is misplaced because errors are not distributed uniformly across tokens: only 5–10% of tokens are "key"—genuinely

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

dependent on long-range context [Fang et al., 2025]—while the remaining majority become nearly deterministic once enough context accumulates. Replacing $(1-e)^n$ with the two-rate model $P(\text{correct}) = (1-e_{\text{key}})^k (1-e_{\text{non}})^{n-k}$ with $k \ll n$ scaling sublinearly in n recovers the observed long-context coherence and converts the reliability question from "how does n grow?" to "how does k grow?".

From sparse tokens to recurring patterns. This paper takes the next step. Sparsity tells us *where* errors live; the natural follow-up is *what* they are. Recent failure-mode atlases supply a striking empirical answer: errors are not only sparse, they are also *repetitive*. ErrorAtlas [Ashury-Tahan et al., 2026], covering 83 models across 35 datasets, organises observed failures into 17 named categories sorted by prevalence, with a long-tailed distribution heavily concentrated in the head. In code, two error types (`AssertionError` and `NameError`) cover 86.35% of failures on HumanEval across 14 LLMs, replicated across 23 models on HumanEval Pro and MBPP Pro [Wen et al., 2024]. In math, MWPES-300K [Sun et al., 2025] categorises 304,865 errors from 15 LLMs across four math word-problem datasets and reports that dataset characteristics shape error patterns systematically. The same picture appears in multi-hop QA [Zhang et al., 2026], agentic tool use [Cemri et al., 2025], and retrieval-augmented generation [Wood and Forbes, 2024]. These observations suggest a third layer in the reliability architecture, complementing key-token sparsity (Layer 1) and within-key stratified-manifold structure (Layer 2 in Anonymous 2025). Within the small set of key tokens, only a fraction β produce *hard* failures, and those hard failures cluster into a finite catalogue of recurring modes whose size grows much more slowly than the number of observed events.

Contributions. The central result is the conditional polylog intervention budget of Proposition 1: $m \geq \lceil |C|^{1-\varepsilon/e_{\text{hard}}} \rceil$, with a doubly-logarithmic rate as the optimistic special case under the postulate above. The *sequence*-level analogue is strictly tighter and approaches full-catalogue coverage as k grows—an asymmetry we surface rather than bury. Around this central claim the paper does four supporting things. The three-layer formal framework—sparsity (α), stratification (β), and bounded mode catalogue ($|C|$)—partitions key tokens into easy and hard subsets, then adds the recurring-pattern catalogue (§3). The empirical postulate of logarithmic mode discovery is calibrated against ErrorAtlas, HumanEval, and MWPES, with $\sigma \in [0.87, 1.85]$ across anchors and $\sigma \approx 1.85$ adopted from ErrorAtlas as the conservative target. Section 4 synthesises evidence for clustering, cluster-selective interventions, and sublinear length scaling, drawing on ~60 prior published results including the six-axis capability-elimination harvest of 28 quantitatively-anchored citations stratified into Patterns A/B/C (Appendix B). And the re-audit of the most-cited steep-decay results (Appendix C) shows each decays over a variable other than raw token length.

2 Related Work

Failure-mode taxonomies. ErrorAtlas [Ashury-Tahan et al., 2026], MWPES-300K [Sun et al., 2025], the HumanEval categorisation of Wen et al. [2024], and the RFMDataset [Guo et al., 2025] together establish that, at any given corpus size, the number of distinct named failure modes is small (typically 8–20) and the cumulative coverage of the top modes is high. Domain-specific taxonomies for multi-hop QA [Zhang et al., 2026], multi-agent systems [Cemri et al., 2025], and tool-augmented agents [Yao et al., 2024] report the same pattern.

Targeted interventions. Each named cluster has been the subject of focused intervention research. Python execution closes most of the arithmetic-error cluster [Gao et al., 2023a, Chen et al., 2023, Gou et al., 2024b]; constrained decoding eliminates format violations [Suresh et al., 2025, Wang et al., 2025b, Dong et al., 2025]; execution feedback closes most code-logic errors [Shinn et al., 2023, Huang et al., 2023]; process reward models reduce step-level reasoning errors [Wang et al., 2024b, Lightman et al., 2024]; RAG strongly reduces hallucinations [Wood and Forbes, 2024]; preference optimisation closes over-refusal [Karaman et al., 2024]; structured uncertainty fixes most tool-call failures [Suri et al., 2025]. Two structural observations recur: each intervention is *cluster-selective* (residuals belong to a different named cluster, not to the targeted one), and *additivity is approximate but not perfect* [Patel et al., 2026, Le, 2026].

Length-decay benchmarks. A parallel literature measures the *shape* of the reliability decay curve. Mild-decay results—Loong [Wang et al., 2024a], GSM-∞ [Zhou et al., 2025], RULER [Hsieh et al., 2024], anchor-based LLMs [Pang et al., 2024]—report log-linear, sigmoidal, or threshold-like decay inconsistent with smooth $(1-\varepsilon)^n$. Steep-decay results [Dziri et al., 2023, Kuratov et al., 2024, Kwa et al., 2025, Wan et al., 2026] are sometimes cited as exponential-compounding evidence. The apparent tension dissolves on careful reading: every steep-decay paper decays over a variable distinct from raw token length, which Appendix C documents.

Gap. What is missing is a quantitative bridge between the small, repeating taxonomy catalogue and a polylog intervention budget. We provide that bridge in §3.

3 Theoretical Framework

Levels of $|C|$. LLM error analysis routinely conflates four distinct objects, which we keep separate. *L1: failure events*—concrete observed errors. *L2: empirical taxonomy categories*—researcher-chosen labels grouping L1 events; what published taxonomies report. *L3: latent failure modes*—the unobserved underlying clusters L2 approximates. *L4: capability axes / interventions*—the engineering unit (Python interpreter, constrained decoder, RAG). Throughout this section, $|C|$ refers to the L2 count, because L2 is what published taxonomies report; Postulate 1 is engineering-relevant because L4 is coarser than L2, and epistemically conditional because L2 is a noisy proxy for L3.

3.1 β-stratification of key tokens

The two-rate model of Anonymous [2025] distinguishes k key tokens (error rate e_{key}) from $n-k$ non-key tokens ($e_{\text{non}} \ll e_{\text{key}}$). Empirical atlases suggest that even within the key-token class errors are not uniformly distributed: most key tokens are "decisions" for which the model has stable representations; only a fraction concentrate the actual failures.

Definition 1 (Hard fraction). Partition the k key tokens of a sequence into easy and hard subsets, $k_{\text{hard}} = \beta k$, $k_{\text{easy}} = (1-\beta)k$, with $\beta \in (0,1)$. Hard key tokens have an elevated error rate e_{hard} corresponding to manifold-transition decisions in the sense of Anonymous [2025, §3.2]; easy key tokens have $e_{\text{easy}} \approx e_{\text{non}}$. The composed reliability becomes

$$P(\text{correct}) = (1-e_{\text{hard}})^{\beta k}(1-e_{\text{easy}})^{(1-\beta)k}(1-e_{\text{non}})^{n-k}.$$

We do not assume iid token errors: the three rates summarise *conditional per-decision failure hazards* after conditioning on prior decisions being correct, and grouping by stratum yields the multiplicative survival expression. Because $e_{\text{easy}} \approx e_{\text{non}}$ once context accumulates, the failure rate is dominated by the βk hard tokens, and we treat e_{hard} as the load-bearing quantity. The parameter β is *latent*: failure-mode atlases report $\Pr(\text{category} \mid \text{error occurred})$, which does not identify $\Pr(\text{hard} \mid \text{key-token decision})$. We carry β symbolically.

3.2 Empirical postulate: logarithmic mode discovery

The βk hard-token failures cluster into a finite set of recurring patterns indexed $c = 1, \ldots, |C|$. The natural question is how $|C|$ grows with the number of observed failures. Two candidates exist in the type–token literature: Heaps' law (power-law, $|C| \approx K \cdot k_{\text{hard}}^b$ with $b \in [0.4, 0.6]$ [Manning et al., 2008]), and logarithmic discovery. **Zipfian rank-frequency does not imply logarithmic growth**: Heaps' law is the correct type–token consequence of Zipf-distributed events, and it is power-law, not logarithmic. We therefore state logarithmic mode discovery as an empirical postulate defensible by direct measurement, not as a theorem.

Two sample-size variables matter and conflating them is the seam that lets reviewers in. Let $h(n) = \beta k(n)$ denote the number of hard-token decisions in a single sequence of length n; let T denote observed hard-failure events in a corpus used to discover the domain catalogue. We write C_D for the full reachable catalogue inside domain patch D, $C_{\text{seen},D}(T)$ for the subset discovered after T failures, and $C_{\text{active},D}(n)$ for the subset a single sequence of length n can activate.

Postulate 1 (Patch-indexed catalogue discovery). *Within a fixed application domain D, the number of named recurring failure modes discovered after T sampled hard-failure events is bounded by*

$$|C_{\text{seen},D}(T)| \le \min\bigl(A_D + \sigma_D \ln T,\ |C_D|\bigr), \qquad \sigma_D > 0,\ A_D \ge 0.$$

The cap $|C_D|$ is the domain-imposed ceiling. The postulate concerns catalogue discovery, not the number of hard decisions inside a single sequence.

Assumption 2 (Per-sequence active-mode exposure). *For a single sequence of length n in domain D,*

$$|C_{\text{active},D}(n)| \le \min\bigl(A'_D + \sigma'_D \ln h(n),\ |C_D|\bigr).$$

Primed constants are distinct from those of Postulate 1; corpus discovery and per-sequence activation need not share the same rate. Proposition 1's sequence-length scaling uses $C_{active,D}(n)$; domain-library budgeting uses C_D.

Empirical defense. ErrorAtlas: $k_{\text{hard}} \gtrsim 10^4$, $|C| = 17$, giving $\sigma \approx 17/\ln(10^4) \approx 1.85$ with $A = 0$. HumanEval [Wen et al., 2024]: $|C| \in [8, 12]$, $\sigma \in [0.87, 1.30]$. MWPES: $|C| \in [15, 20]$, $\sigma \in [1.2, 1.6]$. Across the three anchors, $\sigma \in [0.87, 1.85]$; we carry $\sigma \approx 1.85$ as the conservative target. Endpoint counts at a single corpus scale are not discovery curves; the subsample-discovery measurement remains an explicit empirical test we flag (§6, Appendix A).

Capabilities are coarser than error classes. Postulate 1's $\sigma \approx 1.85$ is calibrated against *named error categories*, but a single capability axis often targets several L2 categories at once. A Python interpreter removes the execution-error component of arithmetic, unit conversion, simple counting, list manipulation, and date arithmetic; a constrained decoder eliminates by construction the format-violation cluster and the structural component of "missing required element." Postulate 1 is therefore a conservative input for the engineering question; the formal coverage problem is weighted set cover, with the ranked-category form of Proposition 1 as a tractable proxy. An alternative power-law (Heaps) formulation and symbolic-form sensitivity analysis across candidate cluster-count laws are given in Appendix A.

Domain patches cap the catalogue. A model deployed in a fixed application domain occupies a bounded sub-region of the stratified manifold—a *domain patch* P_D [Anonymous, 2025, Li and Sarwate, 2025]. Empirical support: embedding-space intrinsic dimensions of ~ 5–15 across domains, well below ambient $\sim 10^3$ [Park et al., 2024, Li and Sarwate, 2025]; cross-domain accuracy spreads of 20–70% on a single base model [Wang et al., 2024c]; log-popularity thresholds below which scaling fails [Mallen et al., 2023, Kandpal et al., 2023]. None of these *measure* $|C_D|$ directly; together they motivate the patch-indexed hypothesis. The patch makes logarithmic or saturating discovery dynamics plausible inside fixed domains; σ_D, A_D, β_D, $|C_D|$ are all domain-indexed.

3.3 Coverage by a targeted intervention library

Given a catalogue of $|C|$ recurring modes, an intervention library of size m covers the top-m, leaving $|C| - m$ uncovered. We adopt the log-coverage form

$$F(m; |C|) = \min\left(1,\ \frac{\ln m}{\ln |C|}\right)$$

as a *postulated* closed expression respecting the $\min(1, \cdot)$ cap. For our ErrorAtlas anchor ($|C| = 17, m = 5$), $F(5; 17) \approx 56.8\%$ and $F(10; 17) \approx 81.3\%$. This is an empirical best-fit choice, not derived from Zipf: ErrorAtlas sorts its 17 categories by prevalence but does not publish a single cross-domain cumulative top-m curve at this writing; alternative cumulatives (Zipf–Mandelbrot, truncated power law, saturating exponential) match the same qualitative shape with comparable freedom. Appendix A shows the polylog conclusion survives across them. Two caveats: (a) F counts named L2 categories covered, not L4 capability axes deployed, so real-deployment residual error is often lower at any given m; (b) we adopt $F(0; |C|) := 0$ and $F(1; |C|) := p_1$ (mass of the most prevalent mode) since the log form is not anchored at $m = 1$. After deploying the top-m library, the residual per-hard-token error rate is $e_{\text{res}}(m) = (1 - F(m; |C|))\, e_{\text{hard}}$.

3.4 Main proposition: conditional polylogarithmic intervention budget

We compose the three ingredients with the two-rate model. For sequence-length scaling we use Assumption 2, not Postulate 1: the empirical sparsity $\alpha = k/n \in [0.05, 0.10]$ is a *finite-length* statement, and under Anonymous [2025, §3.1]'s $k \sim \log n$ regime, α decays as $O(\log n/n)$.

Conditionality. The bound below is engineering math, not a theorem about LLMs. It holds conditional on (i) the log-coverage form of §3.3 on its declared domain $m \ge 2$, $|C_{\text{eff}}| \ge 2$; (ii) the residual target $\varepsilon \in (0, e_{\text{hard}})$; (iii) Assumption 2 for any sequence-length rate claim. A different cumulative fitted to the same anchors preserves the qualitative polylog conclusion (Appendix A) but changes the exponent.

Proposition 1 (Per-Hard-Token Intervention Budget). *Let $\varepsilon \in (0, e_{hard})$ be a target per-hard-token residual error rate for sequences in domain D, with $|C_{eff}| \geq 2$, where $C_{eff} = C_{active,D}(n)$ for per-sequence scaling or $C_{eff} = C_D$ for full domain-library deployment. Under the coverage form of §3.3, the smallest intervention library satisfying $e_{res}(m) \leq \varepsilon$ has size*

$$m \geq \left\lceil |C_{\text{eff}}|^{1-\varepsilon/e_{\text{hard}}} \right\rceil .$$

If $C_{eff} = C_{active,D}(n)$ and $k(n) = \Theta(\log n)$, then in the pre-cap regime $m = O\big((\log \log n)^{1-\varepsilon/e_{hard}}\big)$, i.e., the intervention budget grows doubly logarithmically with sequence length. In the cap regime, or for full domain-library deployment, $m \geq \lceil |C_D|^{1-\varepsilon/e_{hard}} \rceil$, independent of n.

Proof. From §3.3, $e_{\text{res}}(m) = (1 - F(m; |C_{\text{eff}}|))\, e_{\text{hard}}$. Requiring $e_{\text{res}}(m) \leq \varepsilon$ gives $F(m; |C_{\text{eff}}|) \geq 1 - \varepsilon/e_{\text{hard}}$. Since $\varepsilon < e_{\text{hard}}$, the required coverage lies below one, so the cap in F is inactive; with $F = \ln m / \ln |C_{\text{eff}}|$, we obtain $m \geq |C_{\text{eff}}|^{1-\varepsilon/e_{\text{hard}}}$, and taking the ceiling gives the stated form. Assumption 2 gives $|C_{\text{active},D}(n)| = O(\log \log n)$ when $k(n) = \Theta(\log n)$, before the cap. After the cap, $|C_{\text{eff}}| = |C_D|$ and the bound is independent of n. ☐

We label this a *Proposition* rather than a *Theorem* to keep its conditional nature visible. The doubly-logarithmic rate is the *optimistic special case*: per-sequence, pre-cap, under logarithmic mode discovery. Weaker (but still polylog) rates obtain under Heaps (Appendix A); cap regime is a domain-constant.

Sequence-level versus per-hard-token bounds. Proposition 1 bounds the *per-hard-token* residual error. Sequence-level failure probability is strictly stronger. With non-hard-token survival factor $S_{\text{base}} = (1 - e_{\text{easy}})^{(1-\beta)k}(1 - e_{\text{non}})^{n-k}$, requiring $(1 - e_{\text{res}})^{\beta k}\, S_{\text{base}} \geq 1 - \varepsilon_{\text{seq}}$ yields three regimes: (i) if $S_{\text{base}} < 1 - \varepsilon_{\text{seq}}$, the binding constraint is non-key noise and hard-token interventions alone cannot meet the target—this is the back-door route by which the exponential-in-n concern re-enters; (ii) if $S_{\text{base}} \geq 1 - \varepsilon_{\text{seq}}$, the required library size is $m \geq \lceil |C_{\text{eff}}|^{1-\tau_{\text{seq}}/e_{\text{hard}}} \rceil$ with $\tau_{\text{seq}} = 1 - ((1 - \varepsilon_{\text{seq}})/S_{\text{base}})^{1/(\beta k)}$, and as the target tightens toward feasibility, $\tau_{\text{seq}} \to 0$ and $m \to |C_{\text{eff}}|$—essentially every named mode must be covered; (iii) if $\tau_{\text{seq}} \geq e_{\text{hard}}$, the target is met without intervention. The widely-quoted "≈ 50 patterns $\to\approx 90\%$ reduction" claim is a per-hard-token statement; the sequence-level analogue requires correspondingly more interventions. Production systems with one-shot reliability targets should plan for the stricter regime—and check whether they are in Regime (i), where catalogue-budgeting cannot help at all.

4 Empirical Evidence

We collect evidence for three load-bearing claims: errors cluster into a small recurring set (A); each cluster is addressable by one targeted intervention (B); reliability decays sublinearly with output length (C).

Claim A: Failure-mode clustering. The strongest single result is ErrorAtlas [Ashury-Tahan et al., 2026]: 83 models × 35 datasets, ≈ 7,000 failures, 17 named categories with a long-tailed, head-concentrated prevalence ordering. Domain-specific Paretos reproduce the shape inside each domain—math (MWPES top-4 categories dominate per model with strong cross-model overlap [Sun et al., 2025]), code (`AssertionError`+`NameError` = 86.35% on HumanEval, replicated across 23 models [Wen et al., 2024]), multi-hop QA (a single dominant "missing-evidence" mode [Zhang et al., 2026]), agentic tool use (under 20 recurring modes across two studies [Cemri et al., 2025, Yao et al., 2024]), and RAG (a handful of distinguishable hallucination types [Wood and Forbes, 2024]). The catalogue is also *stable* across models: ErrorAtlas reports a fixed category hierarchy across 83 models; RFMDataset [Guo et al., 2025] finds "strikingly similar failure mode distributions" across ten advanced reasoning models; EDIT [Dai et al., 2025] measures a $\approx 4.7\%$ key-step fraction stable across models. Schaeffer et al. [2025] provide the closest theoretical anchor, proving that the observed aggregate power-law scaling of LLM evaluations requires the per-problem success-rate distribution to be heavy-tailed near $p = 0$—structurally related to Postulate 1.

Claim B: Cluster-selective capability interventions. A dedicated harvest yields 28 capability-elimination citations across six independent axes (arithmetic, code execution, format/structure, perception/grounding, knowledge/RAG, verification), each axis independently confirmed by between three and nine citations. The full table and the three structural patterns—Pattern A by-construction, B strong-empirical-with-class-shift, C moderate-with-shift—appear in Appendix B.

Arithmetic is the cleanest case: PAL [Gao et al., 2023a] lifts GSM-Hard from 20.1% to 61.5%, and the residuals are problem-comprehension errors rather than arithmetic ones. Format violations admit something stronger still—constrained decoding zeroes invalid-token probability by construction [Suresh et al., 2025, Wang et al., 2025b, Dong et al., 2025]. Code-logic errors yield to execution feedback (Reflexion + AgentCoder push HumanEval pass@1 from 80% to 96.3%, with residuals in spec-misinterpretation [Shinn et al., 2023, Huang et al., 2023]); reasoning steps to process supervision (Math-Shepherd: Mistral-7B MATH 28.6% $\rightarrow$ 43.5% [Wang et al., 2024b]); hallucinations to RAG, where Acurai [Wood and Forbes, 2024] reports 100% elimination on RAGTruth—a benchmark-conditional empirical result, not a general by-construction guarantee. Two more clusters round out the picture: POROver lifts the not-overrefused rate from 57.6% to 82.1% [Karaman et al., 2024], SAGE-Agent raises When2Call accuracy from 36.5% to 65.2% [Suri et al., 2025].

The class-shift signature—post-intervention residuals belong to structurally different classes—is the empirical content of the cluster-orthogonality claim that underwrites Proposition 1's composition step. Capabilities are coarser than error classes: a single Python interpreter removes execution-error components from five named clusters; constrained decoding eliminates format and the structural component of "missing required element" jointly. A negative control sharpens the selectivity claim. DebugBench [Tian et al., 2024] reports that execution feedback works for syntax and reference errors and is explicitly "unhelpful for logic errors"—exactly where the framework predicts capability provisioning fails.

Claim C: Sublinear length scaling. Direct evidence for sublinear—rather than exponential—decay shows up across very different measurement designs. Loong [Wang et al., 2024a] reports GPT-4o on Chain-of-Reasoning declining $81.6\% \rightarrow 32.9\%$ across the 10K $\rightarrow$ >200K context bins; log-linear over $\log L$, and decisively non-exponential per-token. GSM-∞ [Zhou et al., 2025] attacks the independent-error model from a different angle: exponentially more inference compute yields only *linear* AUC gains, and DeepSeek-R1 maintains $10\%+$ accuracy at 130 reasoning operations where $(1-\varepsilon)^{130}$ for any reasonable ε predicts near-zero. Press et al. [2023] find a $\approx 40\%$ compositionality gap that is approximately *constant* across the GPT-3 family—direct evidence against per-step independent compounding, which would predict the gap to grow with chain length.

Structural correlates of the same phenomenon appear elsewhere. Pang et al. [2024]'s Anchor LLMs achieve $\approx 99\%$ K/V reduction with only minor accuracy compromise: the effective number of distinct attention-key vectors a model needs is far smaller than n. Think-Prune-Train [Costello et al., 2025] raises Pass@1 while Pass@20 plateaus—the manifold-transition signature. RULER [Hsieh et al., 2024] finds threshold behaviour inconsistent with smooth $(1-\varepsilon)^n$. METR [Kwa et al., 2025] fits logistic-in-log-length, which is mathematically sublinear in length itself.

Self-consistency [Wang et al., 2023] is consistent with but does not *prove* clustered structure: Condorcet aggregation of iid Bernoulli chains can yield similar gain magnitudes, so we treat it as suggestive rather than diagnostic. Prominent steep-decay counter-evidence [Dziri et al., 2023, Kuratov et al., 2024, Wan et al., 2026, Karpinska et al., 2024] decays over variables distinct from raw token length; per-paper re-audits showing the relocation pattern are in Appendix C.

5 Practical Implications

Reliability engineering is local patch coverage. Within a fixed domain patch, Proposition 1 makes reliability a small-catalogue engineering problem rather than an asymptotic scaling problem. A team should initially budget for a library on the order of tens of interventions, then refine from the local mode-discovery curve $C_{\text{seen},D}(T)$ and the empirical rank-frequency distribution. Available evidence is consistent with ≈ 50 named interventions covering the head of the per-hard-token failure distribution in many measured domains, but this number is a planning prior, not a universal constant: the same base model in cardiology RAG, legal contract drafting, and code review yields three different libraries because C_D, A_D, σ_D all change with the deployment domain.

Per-hard-token vs. sequence-level targets matter for SLA design. Per-hard-token residual error—the right metric for continuous-correction systems—has a polylog budget. Sequence-level failure probability—the right metric for one-shot systems—is strictly stricter and approaches full-catalogue coverage as k grows. Production deployments should design SLAs to match the actual cost structure, not read the per-token result as the production SLA.

Library design as engineering, not asymptotic theory. An intervention library can be assembled module-by-module, in any order, as long as modules target distinct failure classes. Layer-separated interventions (constrained decoding + retrieval + process supervision + tool call) compose approximately additively in our data; the Le [2026] caveat applies only to interventions sharing a prompt channel. Library construction is therefore highly parallelisable.

Capability libraries are coarser than error-class libraries. Five named math classes (arithmetic, units, counting, list manipulation, date arithmetic) collapse under one Python interpreter; three code classes (SyntaxError, NameError, most TypeError) collapse under one execution-feedback loop; format violations and the structural component of "missing required element"—together more than 20% of ErrorAtlas—collapse under one constrained-decoder deployment. A team budgeting for ≈ 50 interventions per domain can expect to need fewer than 50 capability-axis interventions while covering ≈ 50 named clusters. The six axes of Appendix B are closer to the right unit of engineering accounting than the twelve clusters.

6 Discussion

By-construction elimination as a boundary case. Seven of the 28 capability-elimination citations (Appendix B; constrained decoders [Suresh et al., 2025, Dong et al., 2025, Li et al., 2026, Zhang et al., 2023, OpenAI, 2024], proof kernels [Ren et al., 2025], static syntax checks) achieve residual error rate equal to zero *by mathematical construction*, not statistically. Constrained decoders set $P(\text{invalid token}) = 0$ at every generation step; the class of grammar-violating outputs is mathematically empty. In this regime, the polylog bound of Proposition 1 is loose: covered clusters contribute zero to residual error, not a small positive quantity. This is a strengthening result, not a counter-example—capability provisioning erases the error class entirely. Pattern A applies precisely to structural/verifiable classes (format, syntax, schema, invalid proofs), which are also the head of the heavy-tailed cluster frequency distribution. The strongest mechanism applies exactly where the catalogue is densest, compounding the polylog conclusion.

Counter-evidence relocates, not dissolves. Five prominent steep-decay papers [Dziri et al., 2023, Kuratov et al., 2024, Kwa et al., 2025, Wan et al., 2026, Karpinska et al., 2024] are routinely cited as exponential-compounding evidence. Every one decays over a variable distinct from raw token length: compositional graph size [Dziri et al., 2023], number of supporting facts [Kuratov et al., 2024], log-time horizon [Kwa et al., 2025], capacity threshold [Wan et al., 2026], evidence scope [Karpinska et al., 2024]. The apparent rapid decay is, in each case, in a quantity our framework already concentrates the action in (k_{hard} and $|C|$), not in raw sequence length—a relocation, not a dissolution. These regimes remain hard; the framework's value is directing intervention toward capability provisioning along the actual decay axis rather than toward context-window expansion that does not help. Appendix C walks through each paper.

Limitations. Postulate 1 is empirical, not derived; a domain with genuinely Heaps-power-law mode discovery would invalidate the doubly-logarithmic special case while preserving the qualitative polylog conclusion. The coverage form of §3.3 is an empirical best-fit; readers should not interpret the numerical match against anchors as theoretical confirmation. Cluster-orthogonality is approximate; Le [2026]'s caveat on prompt-channel interference tightens the bound in shared-channel settings. Domain narrowness: Postulate 1's empirical anchor rests on three taxonomies (ErrorAtlas, HumanEval, MWPES), all published 2025–2026 and covering general/code/math; the framework is untested on agentic workflows, long-running scientific reasoning, and multi-turn tool use over millions of tokens—precisely the regimes where $|C|$ may grow faster than logarithmically. The $\sigma \approx 1.85$ estimate is a single-point calibration, not a discovery-curve fit; a subsample-vs-distinct-modes plot has not been published for any LLM error taxonomy at this writing. β is latent and we report no empirical range. Taxonomy granularity (the L2–L3 gap) is unmeasured. Patch-shift between deployments changes $A_D, \sigma_D, \beta_D, |C_D|$ simultaneously; libraries calibrated against one patch under-cover the next. Finally, the framework *relocates* the difficulty of long-context reliability rather than resolving it: where k_{hard} grows with task length (adversarial compositional structure, multi-hop chains, long agent horizons), reliability remains hard. The contribution is to name the on-axis intervention, not to make those regimes easy. A falsifiability test the framework passes: DebugBench [Tian et al., 2024] explicitly catalogues which classes execution-feedback can and cannot repair, and the empirical split (works for syntax/reference; "unhelpful for logic errors") is exactly what the framework's selectivity claim predicts.

7 Conclusion

LLM reliability is often framed as an asymptotic scaling problem. Prior work in this line [Anonymous, 2025] argued that the framing rests on a false uniformity assumption—errors concentrate at ≈ 5–10% of tokens. This paper takes the next step: within that sparse set, errors are not only concentrated but also *repetitive*, clustering into a finite catalogue whose size, under the empirical postulate of §3.2, grows logarithmically—and under the conservative Heaps alternative (Appendix A), as a small power—of observed failures. The conditional consequence (Proposition 1) is that the intervention budget required to bound per-hard-token residual error scales polylogarithmically in sequence length within a fixed domain patch, and becomes a domain-constant once the patch ceiling $|C_D|$ is reached. Sequence-level reliability targets are strictly tighter. Available evidence is consistent with libraries on the order of tens of interventions covering the head of the per-hard-token failure distribution in many fixed domains. The exact number is patch-indexed and should be estimated from local discovery and rank-coverage curves, not assumed from cross-domain priors. This reframes the question from "can we bound the growth of n-token error?" to "have we catalogued enough failure modes?" The latter is finite and addressable. Direct empirical measurement of the mode-rate constant σ on new domains—agentic, scientific, code-in-production—is the open empirical question this paper invites; future work flagged by Anonymous [2025, §6] on attention-mechanism bounds and stratified-manifold geometry would replace the empirical postulate with a derived result.

References

Anonymous. Beyond exponential decay: Rethinking error accumulation in large language models. *arXiv preprint*, 2025. Author field anonymized for double-blind submission; arXiv ID public.

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations (ICLR)*, 2024.

Shir Ashury-Tahan, Yifan Mai, Elron Bandel, Michal Shmueli-Scheuer, and Leshem Choshen. ErrorMap and ErrorAtlas: Charting the failure landscape of large language models. *arXiv preprint*, 2026.

Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya Parameswaran, Dan Klein, Kannan Ramchandran, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. Why do multi-agent LLM systems fail? *arXiv preprint*, 2025. Introduces the MAST taxonomy (Multi-Agent System Failure Taxonomy).

Wenhu Chen, Xueguang Ma, Xinyi Wang, and William W. Cohen. Program of thoughts prompting: Disentangling computation from reasoning for numerical reasoning tasks. *Transactions on Machine Learning Research (TMLR)*, 2023. Cited as 2022 in body.

Kanzhi Cheng, Qiushi Sun, Yougang Chu, Fangzhi Xu, Yantao Li, Jianbing Zhang, and Zhiyong Wu. SeeClick: Harnessing GUI grounding for advanced visual GUI agents. In *Proceedings of ACL*, 2024.

Caia Costello, Simon Guo, Anna Goldie, and Azalia Mirhoseini. Think, prune, train, improve: Scaling reasoning without scaling models. *arXiv preprint*, 2025.

Chengwei Dai et al. Capture the key in reasoning to enhance CoT distillation generalization. In *Proceedings of ACL*, 2025. Earlier arXiv version titled "Beyond Imitation: Learning Key Reasoning Steps from Dual Chain-of-Thoughts in Reasoning Distillation".

Alex Dantart. Reliability by design: Quantifying and eliminating fabrication risk in LLMs. from generative to consultative AI: A comparative analysis in the legal domain and lessons for high-stakes knowledge bases. *arXiv preprint*, 2026.

Yixin Dong, Charlie F. Ruan, Yaxing Cai, Ruihang Lai, Ziyi Xu, Yilong Zhao, and Tianqi Chen. XGrammar: Flexible and efficient structured generation engine for large language models. In *Proceedings of MLSys*, 2025.

Nouha Dziri, Ximing Lu, Melanie Sclar, Xiang Lorraine Li, Liwei Jiang, Bill Yuchen Lin, Peter West, Chandra Bhagavatula, Ronan Le Bras, Jena D. Hwang, et al. Faith and fate: Limits of transformers on compositionality. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Lizhe Fang, Yifei Wang, Zhaoyang Liu, Chenyang Zhang, Stefanie Jegelka, Jianfeng Gao, Bolin Ding, and Yisen Wang. What is wrong with perplexity for long-context language modeling? In *International Conference on Learning Representations (ICLR)*, 2025.

Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. PAL: Program-aided language models. In *Proceedings of ICML*, 2023a. Cited as 2022 in body.

Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. Enabling large language models to generate text with citations. In *Proceedings of EMNLP*, 2023b. Introduces the ALCE benchmark.

Alex J. Goodell, Simon N. Chu, Dara Rouholiman, and Larry F. Chu. Large language model agents can use tools to perform clinical calculations. *npj Digital Medicine*, 8(1):163, 2025. doi: 10.1038/s41746-025-01475-8.

Boyu Gou, Ruohan Wang, Boyuan Zheng, Yanan Xie, Cheng Chang, Yiheng Shu, Huan Sun, and Yu Su. Navigating the digital world as humans do: Universal visual grounding for GUI agents. In *International Conference on Learning Representations (ICLR)*, 2025. ICLR 2025 Oral. Introduces the UGround model.

Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Nan Duan, and Weizhu Chen. CRITIC: Large language models can self-correct with tool-interactive critiquing. In *International Conference on Learning Representations (ICLR)*, 2024a.

Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Minlie Huang, Nan Duan, and Weizhu Chen. ToRA: A tool-integrated reasoning agent for mathematical problem solving. In *International Conference on Learning Representations (ICLR)*, 2024b. Cited as 2023 in body.

Dadi Guo, Jiayu Liu, Zhiyuan Fan, Zhitao He, Haoran Li, Yuxin Li, Yumeng Wang, and Yi R. Fung. Mathematical proof as a litmus test: Revealing failure modes of advanced large reasoning models. *arXiv preprint*, 2025. Introduces the RFMDataset (Reveal Failure Modes).

Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekesh, Fei Jia, Yang Zhang, and Boris Ginsburg. RULER: What's the real context size of your long-context language models? In *Conference on Language Modeling (COLM)*, 2024.

Dong Huang, Qingwen Bu, Jie M. Zhang, Michael Luck, and Heming Cui. AgentCoder: Multi-agent-based code generation with iterative testing and optimisation. *arXiv preprint*, 2023.

Nikhil Kandpal, Haikang Deng, Adam Roberts, Eric Wallace, and Colin Raffel. Large language models struggle to learn long-tail knowledge. In *Proceedings of ICML*, 2023.

Batuhan K. Karaman, Ishmam Zabir, Alon Benhaim, Vishrav Chaudhary, Mert R. Sabuncu, and Xia Song. POROver: Improving safety and reducing overrefusal in large language models with overgeneration and preference optimization. *arXiv preprint*, 2024.

Marzena Karpinska, Katherine Thai, Kyle Lo, Tanya Goyal, and Mohit Iyyer. One thousand and one pairs: A "novel" challenge for long-context language models. In *Proceedings of EMNLP*, 2024.

Yuri Kuratov, Aydar Bulatov, Pavel Anokhin, Ivan Rodkin, Dmitry Sorokin, Artyom Sorokin, and Mikhail Burtsev. BABILong: Testing the limits of LLMs with long context reasoning-in-a-haystack. In *NeurIPS Datasets and Benchmarks Track*, 2024.

Thomas Kwa et al. Measuring AI ability to complete long software tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. METR technical report.

Yifan Le. Schema key wording as an instruction channel in structured generation under constrained decoding. *arXiv preprint*, 2026.

Yann LeCun. Auto-regressive LLMs are doomed. Tweet, March 26, 2023. `https://x.com/ylecun/status/1640122342570336267`, 2023.

Linzhang Li, Yixin Dong, Guanjie Wang, Ziyi Xu, Alexander Jiang, and Tianqi Chen. XGrammar-2: Efficient dynamic structured generation engine for agentic LLMs. *arXiv preprint*, 2026.

Xuyang Li and Anand D. Sarwate. Unraveling the localized latents: Learning stratified manifold structures in LLM embedding space with sparse mixture-of-experts. *arXiv preprint*, 2025.

Yujia Li et al. Competition-level code generation with AlphaCode. *Science*, 378(6624):1092–1097, 2022. doi: 10.1126/science.abq1158.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step. In *International Conference on Learning Representations (ICLR)*, 2024. Cited as 2023 in body.

Fangyu Liu, Julian Martin Eisenschlos, Francesco Piccinno, Syrine Krichene, Chenxi Pang, Kenton Lee, Mandar Joshi, Wenhu Chen, Nigel Collier, and Yasemin Altun. DePlot: One-shot visual language reasoning by plot-to-table translation. In *Findings of ACL*, 2023.

Yadong Lu, Jianwei Yang, Yelong Shen, and Ahmed Awadallah. OmniParser for pure vision based GUI agent. *arXiv preprint*, 2024.

Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *Proceedings of ACL*, 2023. Introduces the PopQA benchmark.

Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. *Introduction to Information Retrieval*. Cambridge University Press, 2008. Section on Heaps' law: `https://nlp.stanford.edu/IR-book/html/htmledition/heaps-law-estimating-the-number-of-terms-1.html`.

Ayana Niwa and Hayate Iso. AmbigNLG: Addressing task ambiguity in instruction for NLG. In *Proceedings of EMNLP*, 2024.

OpenAI. Introducing structured outputs in the API. OpenAI Engineering Blog. `https://openai.com/index/introducing-structured-outputs-in-the-api/`, 2024. Published August 6, 2024.

OpenAI. GPT-5 system card. `https://cdn.openai.com/gpt-5-system-card.pdf`, 2025. Published August 7, 2025.

Jianhui Pang, Fanghua Ye, Derek F. Wong, Xun He, Wenxiang Chen, and Longyue Wang. Anchor-based large language models. In *Findings of ACL*, 2024.

Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. In *Proceedings of ICML*, 2024.

Khush Patel, Siva Surendira, Jithin George, and Shreyas Kapale. The six sigma agent: Achieving enterprise-grade reliability in LLM systems through consensus-driven decomposed execution. *arXiv preprint*, 2026.

Ofir Press, Muru Zhang, Sewon Min, Ludwig Schmidt, Noah A. Smith, and Mike Lewis. Measuring and narrowing the compositionality gap in language models. In *Findings of EMNLP*, 2023.

Z. Z. Ren et al. DeepSeek-Prover-V2: Advancing formal mathematical reasoning via reinforcement learning for subgoal decomposition. *arXiv preprint*, 2025.

Rylan Schaeffer et al. How do large language monkeys get their power (laws)? In *International Conference on Machine Learning (ICML)*, 2025.

Yu Shang, Yu Li, Keyu Zhao, Likai Ma, Jiahe Liu, Fengli Xu, and Yong Li. AgentSquare: Automatic LLM agent search in modular design space. *arXiv preprint*, 2024.

Yuling Shi, Songsong Wang, Chengcheng Wan, Min Wang, and Xiaodong Gu. From code to correctness: Closing the last mile of code generation with hierarchical debugging. *arXiv preprint*, 2024. Introduces the MGDebugger method.

Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Yuhong Sun, Zhangyue Yin, Xuanjing Huang, Xipeng Qiu, and Hui Zhao. Error classification of large language models on math word problems: A dynamically adaptive framework. In *arXiv preprint*, 2025. Introduces the MWPES-300K dataset (304,865 error samples from 15 LLMs across four MWP datasets).

Tarun Suresh et al. DINGO: Constrained inference for diffusion LLMs. *arXiv preprint*, 2025.

Manan Suri, Puneet Mathur, Nedim Lipka, Franck Dernoncourt, Ryan A. Rossi, and Dinesh Manocha. Structured uncertainty guided clarification for LLM agents. *arXiv preprint*, 2025. Introduces the SAGE-Agent method.

Runchu Tian, Yining Ye, Yujia Qin, Xin Cong, Yankai Lin, Zhiyuan Liu, and Maosong Sun. DebugBench: Evaluating debugging capability of large language models. In *Findings of ACL*, 2024.

Akihiko Wada et al. Retrieval-augmented generation elevates local LLM quality in radiology contrast media consultation. *npj Digital Medicine*, 8(1):395, 2025. doi: 10.1038/s41746-025-01802-z.

Kai Wan et al. A Fano-style accuracy upper bound for LLM single-pass reasoning in multi-hop QA. In *International Conference on Learning Representations (ICLR)*, 2026. Cited as 2025 in body.

Benlu Wang et al. From scores to steps: Diagnosing and improving LLM performance in evidence-based medical calculations. In *Proceedings of EMNLP*, 2025a. Introduces the MedRaC benchmark.

Darren Yow-Bang Wang, Zhengyuan Shen, Soumya Smruti Mishra, Zhichao Xu, Yifei Teng, and Haibo Ding. SLOT: Structuring the output of large language models. *arXiv preprint*, 2025b. SLOT = Structured LLM Output Transformer.

Minzheng Wang et al. Leave no document behind: Benchmarking long-context LLMs with extended multi-doc QA. In *Proceedings of EMNLP*, 2024a.

Peiyi Wang, Lei Li, Zhihong Shao, R. X. Xu, Damai Dai, Yifei Li, Deli Chen, Y. Wu, and Zhifang Sui. Math-Shepherd: Verify and reinforce LLMs step-by-step without human annotations. In *Proceedings of ACL*, 2024b. Cited as 2023 in body.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

Yubo Wang et al. MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. In *NeurIPS Datasets and Benchmarks Track (Spotlight)*, 2024c.

Hao Wen et al. Fixing function-level code generation errors for foundation large language models. *arXiv preprint*, 2024. Introduces the LlmFix method. Reports AssertionError 63.64% + NameError 22.71% = 86.35% of HumanEval failures across 14 LLMs.

Michael C. Wood and Adam A. Forbes. 100% elimination of hallucinations on RAGTruth for GPT-4 and GPT-3.5 Turbo. *arXiv preprint*, 2024. Introduces the Acurai method.

Tianbao Xie et al. Scaling computer-use grounding via user interface decomposition and synthesis. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. NeurIPS 2025 Spotlight. Introduces the Jedi grounding dataset and OSWorld-G benchmark.

Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ-bench: A benchmark for tool-agent-user interaction in real-world domains. *arXiv preprint*, 2024.

Cyril Zakka, Rohan Shad, Akash Chaurasia, Alex R. Dalal, Jennifer L. Kim, Michael Moor, Robyn Fong, Curran Phillips, Kevin Alexander, Euan Ashley, Jack Boyd, Kathleen Boyd, Karen Hirsch, Curtis Langlotz, Rita Lee, Joanna Melia, Joanna Nelson, Karim Sallam, Stacey Tullis, Melissa Ann Vogelsong, John Patrick Cunningham, and William Hiesinger. Almanac — retrieval-augmented language models for clinical medicine. *NEJM AI*, 1(2):AIoa2300068, 2024. doi: 10.1056/AIoa2300068.

Kexun Zhang et al. Don't fine-tune, decode: Syntax error-free tool use via constrained decoding. *arXiv preprint*, 2023. Introduces the ToolDec method.

Meiru Zhang, Zaiqiao Meng, and Nigel Collier. Failure modes in multi-hop QA: The weakest link effect and the recognition bottleneck. *arXiv preprint*, 2026.

Yang Zhou, Hongyi Liu, Zhuoyan Chen, Yuandong Tian, and Beidi Chen. GSM-Infinite: How do your LLMs behave over infinitely increasing context length and reasoning complexity? *arXiv preprint*, 2025.

A Heaps Power-Law Variant and Cluster-Count Sensitivity

A reader who prefers to derive cluster-count growth from standard Heaps' law rather than from Postulate 1 obtains a qualitatively similar result. Let $|C|(k_{\text{hard}}) \approx K \cdot k_{\text{hard}}^b$ with $b \in (0, 1)$. Canonical fits give $b \approx 0.5$ for natural-language vocabularies; failure-mode taxonomies plateau much more sharply (ErrorAtlas stabilises at $|C| = 17$ across 10^4+ failures), implying a smaller effective $b \approx 0.2$ in our setting. Composing with $k = \Theta(\log n)$, we get $|C| = O((\log n)^b)$, and Proposition 1 becomes

$$m = O\Big((\log n)^{b\cdot(1-\varepsilon/e_{\text{hard}})}\Big).$$

This is still polylogarithmic in n for any $b \in (0, 1)$ and any $\varepsilon < e_{\text{hard}}$. The paper's qualitative claim survives either choice of cluster-count law.

Symbolic-form sensitivity across candidate laws. The polylog conclusion depends on which cluster-count law one accepts; available evidence is consistent with multiple candidates because no subsample-discovery curve has been published for any LLM failure-mode taxonomy at this writing. We therefore report symbolic rates rather than fitted constants. With $h(n) = \beta k(n)$:

- **Logarithmic:** $|C_{\text{active},D}(n)| = O(\log h(n))$. If $k(n) = \Theta(\log n)$, then $m = O\big((\log\log n)^{1-\varepsilon/e_{\text{hard}}}\big)$.
- **Heaps:** $|C_{\text{active},D}(n)| = O\big(h(n)^b\big)$ with $b \in (0, 1)$. If $k(n) = \Theta(\log n)$, then $m = O\big((\log n)^{b\,(1-\varepsilon/e_{\text{hard}})}\big)$.
- **Saturating:** $|C_{\text{active},D}(n)| \le |C_D|$. Then $m = O\big(|C_D|^{1-\varepsilon/e_{\text{hard}}}\big)$, constant in n once the patch ceiling is reached.

The qualitative conclusion is robust **under the** $k(n) = \Theta(\log n)$ **regime**: m grows more slowly than any positive power of n under every candidate, and the directional claim ("a small library covers the head of the failure distribution in the per-hard-token regime") survives. Only the exponent shifts: doubly-logarithmic under logarithmic discovery, $(\log n)^b$ with small b under Heaps, constant in the cap regime. The doubly-logarithmic rate is the optimistic special case. If $k(n)$ grows as a positive power of n, the Heaps variant inherits that power and the polylog-in-n language fails along that axis—the framework's intervention prescription still applies, but its asymptotic-rate framing does not.

Numerical constants require a measured discovery curve $C_{\text{seen},D}(T)$ or $C_{\text{active},D}(n)$. Existing taxonomies provide endpoint category counts at a single corpus scale (ErrorAtlas at $|C| = 17$ for

$\approx 10^4$ failures), not discovery curves. The subsample-discovery measurement remains the explicit empirical test that would either tighten the postulate or fall back to the Heaps variant. Until that measurement exists, the framework's headline rate should be read as *polylogarithmic in the pre-cap regime, domain-constant in the cap regime*—with the specific exponent flagged as a falsifiability test rather than a fitted prediction.

B Full Failure-Mode Taxonomy and the Capability-Elimination Harvest

The intervention literature provides at least one targeted countermeasure for each named cluster of §4. Two organising granularities exist: at the capability level (six axes) the cluster-orthogonality property is most clearly visible; at the error-class level (twelve named clusters) the evidence aligns with the taxonomies of §4 but is finer-grained than the underlying capability mechanisms.

Six capability axes and three structural patterns. A dedicated harvest yields **28 quantitatively-anchored citations** across **six independent capability axes**: Arithmetic (Python/symbolic execution), Code Execution (REPL/sandbox feedback), Format/Structure (constrained decoding, FSMs, grammar engines), Perception/Grounding (visual grounding for GUI, charts, tables), Knowledge/RAG (dense retrieval and citation grounding), Verification (proof checkers, learned verifiers, classifier rerouting, process supervision). Each axis is independently confirmed by between three and nine citations. We stratify the 28 by kind of evidence into three patterns:

Pattern A — Hard guarantees (by-construction). Seven citations achieve residual error rate equal to zero *by construction*, restricted strictly to structural/verifiable classes: constrained decoders set $P(\text{invalid token}) = 0$ at every step [Suresh et al., 2025, Zhang et al., 2023, Dong et al., 2025, Li et al., 2026, OpenAI, 2024]; static syntax checks reject programs with a `SyntaxError` before execution [Wen et al., 2024]; proof kernels reject any output failing type-checking [Ren et al., 2025]. The class of grammar-violating outputs is mathematically empty under these mechanisms.

Pattern B — Strong empirical reductions with class-shift signature. Roughly fourteen citations report empirical reductions of 80–100% in a named error class, with the post-intervention failure log dominated by structurally different residual classes. Program-of-Thoughts on GSM8K [Chen et al., 2023]: calculation errors drop from 30% of failures to 0%, residuals are 62% reasoning + 36% misunderstanding. OpenMedCalc [Goodell et al., 2025]: "only interpretation errors were identified." Acurai [Wood and Forbes, 2024]: 100% hallucination elimination on RAGTruth (95% CI 91–100%)—strong empirical, not by-construction.

Pattern C — Moderate reductions (60–80%) with residuals shifting outside the target class. Legal RAG [Dantart, 2026]: fabricated citations $> 30\% \rightarrow < 0.2\%$. GPT-5 SimpleQA with web access [OpenAI, 2025]: $47\% \rightarrow 9.6\%$ (inter-condition, not within-condition). CRITIC [Gou et al., 2024a]: toxic generation -79.2%.

The 28 citations, organised by axis and pattern.

Axis	Citation & targeted class	Pre → Post	Pattern
Verification	Ren et al. [2025] — Invalid Lean proof	any → 0% by constr.	A
Format	OpenAI [2024] — JSON schema violation	>60% → 0% by constr.	A
Format	Zhang et al. [2023] — Tool-call syntax	21–100% → 0%	A
Format	Suresh et al. [2025] — JSON parse failure	13–82% → 0%	A
Format	Dong et al. [2025] — Multi-format errors	20–38% → 0%	A
Format	Li et al. [2026] — Malformed tool calls	33–78% → 0%	A
Arithmetic	Chen et al. [2023] — Calculation on GSM8K	30% → 0% of failures	B
Arithmetic	Goodell et al. [2025] — Clinical arithmetic	"only interpretation errors"	B
Knowledge/RAG	Wada et al. [2025] — RAG hallucinations	8% → 0% ($p = 0.012$)	B
Knowledge/RAG	Wood and Forbes [2024] — Context-conflict hallucinations	100% → 0% on subset	B
Knowledge/RAG	Dantart [2026] — Fabricated legal citations	>30% → <0.2%	C
Code Exec	Wen et al. [2024] — `SyntaxError` (HumanEval)	5.76% → 0.01%	A
Code Exec	Li et al. [2022] — False-positive submissions	62% → 4%	B
Code Exec	Shi et al. [2024] — 5 code-bug classes	100% repair (5/6 classes)	B
Knowledge/RAG	Gao et al. [2023b] — Citation grounding (ELI5)	≈50% w/o complete support	C
Knowledge/RAG	OpenAI [2025] — Factual errors	47% → 9.6% w/web	C
Knowledge/RAG	Zakka et al. [2024] — Clinical citation errors	44% → 9%	C
Arithmetic	Wang et al. [2025a] — Arithmetic in medical reasoning	426 → 74 errors	B
Code Exec	Wen et al. [2024] — `NameError` repair	22.7% → 2.3%	C
Perception	Gou et al. [2025] — GUI grounding errors	16.2% → 73.3% acc.	B
Perception	Xie et al. [2025] — GUI task failures	5% → 27% SR (5.4×)	B
Perception	Cheng et al. [2024] — Element mislocation	5.2% → 53.4% (10.3×)	B
Perception	Liu et al. [2023] — Chart-reading errors	38.2% → 67.6% (+29.4 pp)	B
Verification	Gou et al. [2024a] — Toxic generation	−79.2%	C
Verification	Wang et al. [2024b] — Reasoning step errors	28.6% → 43.5% w/rerank	B
Knowledge/RAG	Asai et al. [2024] — Unsupported facts	55.5% → 22% error	B
Perception	Lu et al. [2024] — Icon/element errors	70.5% → 93.8% (79% red.)	B
Knowledge/RAG	Mallen et al. [2023] — Long-tail entity errors	≈80% → ≈50% failure	B

Capabilities are coarser than error classes. Several entries in the harvest reveal "two-for-one" reductions where one capability addresses multiple named clusters: a single Python interpreter removes the execution-error component of arithmetic, unit conversion, simple counting, list manipulation, and date arithmetic; constrained decoding eliminates by construction both format violations and the structural component of "missing required element"; code execution feedback strongly reduces `SyntaxError`, `NameError`, and most `TypeError` together; RAG strongly reduces factual hallucinations, fabricated citations, and outdated information jointly. The practical capability library required is therefore smaller than the count of named error categories.

The twelve named clusters.

Cluster	Axis	Intervention	Before → After	Patt.
A. Arithmetic	Arithmetic	PAL Python [Gao et al., 2023a]	20.1% → 61.5%	B
B. Unit conversion	Arithmetic	Folded into A (Python)	—	—
C. Counting	Arithmetic	Explicit counter (gap)	—	gap
D. Format/schema	Format	DINGO/SLOT [Suresh et al., 2025, Wang et al., 2025b]	up to +68 pp; 99.5% acc.	A
E. Code logic/type	Code Exec	Reflexion/AgentCoder [Shinn et al., 2023, Huang et al., 2023]	80% → 96.3% pass@1	A/B
F. Multi-hop drift	Knowledge	Entity-grounded rewriter	≈5–10 pp	C
G. Reasoning step	Verification	Math-Shepherd [Wang et al., 2024b]	28.6% → 43.5%	B
H. Spec misinterpret.	Verification	Clarification [Niwa and Iso, 2024]	ROUGE-L +15	C
I.a Struct. missing	Format	Subsumed by D (constr. decode)	—	A
I.b Semantic missing	Verification	Subsumed by G/H	—	C
J. Hallucination	Knowledge	RAG [Wood and Forbes, 2024]	non-zero → 0%	B/C
K. Refusal	Verification	POROver [Karaman et al., 2024]	57.6% → 82.1%	B
L. Tool/API	Format+Verif.	SAGE-Agent [Suri et al., 2025]	36.5% → 65.2%	B

Eight of twelve categories have a strong citation with double-digit percentage-point improvement; three (B, C, I) lack a clean single-cluster ablation; one (F) has consistent medium-strength evidence. Category I (missing required elements) splits mechanistically into I.a (structural, absorbed by D's constrained decoders) and I.b (semantic, absorbed by G/H). Categories B (units) and C (counting) remain technical gaps: B is naturally folded into A; C is the smallest residual gap.

Additivity and its limits. Patel et al. [2026] demonstrate that stacking interventions targeting orthogonal failure modes can produce large compound reliability gains: their parallel-consensus framework yields a $14{,}700\times$ improvement over single-pass baseline—evidence for compound gains from decomposition plus consensus aggregation under their specific parallel-voting regime, not direct demonstration that heterogeneous cluster-targeted interventions compose additively without voting. Shang et al. [2024] similarly show supra-additive gains when modules target distinct failure modes. Le [2026] reports that schema-level and prompt-level instructions interact *non-additively* when sharing a prompt channel: interventions on orthogonal processing layers (decoding constraint vs. retrieval vs. training-signal vs. inference-time tool call) compose near-additively, while those sharing a channel may interfere.

The irreducible-semantic residual. Of the 17 ErrorAtlas categories, 13 are addressed under Patterns A, B, or C by one of the six capability axes; four are not—residuals of inappropriate refusal beyond preference optimisation, specification misinterpretation not closed by clarification, problem-decomposition reasoning bottlenecks, and a "user wanted something different" semantic remainder. These are classes where the failure is in choosing what to do, not executing it. Proposition 1's prediction of $O(\log)$ rather than $O(0)$ residual reliability reflects exactly this irreducible core. The framework does not claim 100% coverage of all failures; it claims polylog-bounded *capability-eliminable* failures, with the named semantic residual as the floor.

C Counter-Evidence Re-Audits

Five prominent papers are routinely cited as evidence that LLM reliability decays steeply with length [Dziri et al., 2023, Kuratov et al., 2024, Kwa et al., 2025, Wan et al., 2026, Karpinska et al., 2024]. A careful re-reading shows that *every one* of these papers decays over a variable distinct from raw token length n. Identifying the decay axis is not the same as dissolving the practical concern: where compositional graph size, fact count, or evidence scope grow with problem length, the framework predicts steep failure curves. The contribution is to identify which interventions help (capability provisioning along the actual decay axis) and which do not.

Dziri et al. [2023], Faith and Fate. GPT-4 multi-digit multiplication accuracy drops from 59% (3-digit) to 4% (4-digit) zero-shot, with the authors theorising "probability of incorrect predictions converges exponentially to ≈ 1 for abstract compositional tasks." The decay variable is *compositional graph size* N, not raw token length: the 3×3 graph has on the order of d^2 partial products plus carries, and multi-digit multiplication is engineered so that $k_{\text{hard}} \approx N$—every node in the computation graph is a hard decision. For natural-language tasks where $k_{\text{hard}} \ll n$, Dziri's regime is the *boundary case* in which our framework reduces to their result. A clean $(1-\varepsilon)^N$ exponential cannot simultaneously reproduce 59% at 3-digit and 4% at 4-digit for any single per-node ε—the observed drop is locally steeper than per-node iid exponential, consistent with a finite catalogue of failure modes exhausting as N grows. *Where the framework concedes*: in adversarial compositional tasks engineered so $k_{\text{hard}} \approx N$, the framework reduces to Dziri's regime and does not relieve it; the relocation is informational, not magical.

Kuratov et al. [2024]. The abstract reports "performance declines sharply with increased reasoning complexity" and models "effectively utilise only 10–20% of the context." Read in detail, BABILong varies two axes: context length n (0K to 10M tokens) and number of supporting facts k (QA1 $= 1$ fact to QA3 $= 3$ facts). The sharp decline is in k, not n. For QA1 (single-fact), most models "perform well up to 4,000 tokens"—a plateau, not exponential decay. The famous RAG-flat-across-length result (60% on single-fact QA *independent of context length*) is the cleanest possible demonstration that when relevant evidence is in window, length does not matter. Recurrent Memory Transformers maintaining performance to 50M tokens further confirms effective k is determined by architecture, not raw n. The framework's concession is narrow but real: for multi-hop tasks whose required fact count grows with task complexity, BABILong's sharp k-axis decay is exactly what the framework predicts happens, not a counter-example.

Kwa et al. [2025] (METR). Per-model success fits a logistic in $\log$(human-task-duration): $S(t) = \sigma(\beta(\log t - \log h))$. Logistic-in-log-length is *mathematically sublinear* in length itself. The 80%-horizon being 4–$6\times$ shorter than the 50%-horizon is a steep within-model cliff but compatible by construction with the polylog result—a cliff at a specific capacity threshold is a manifold-transition signature. The famous exponential—capability doubling every seven months—is an *inter-model* claim about how the horizon h moves across generations, orthogonal to within-model decay shape. An honest qualifier: the within-model logistic-in-log cliff is steep on a practical scale. Calling it "sublinear in length" is technically correct but engineering-useful only for systems designed to operate well below the 50% horizon.

Wan et al. [2026], Fano-style upper bound. This paper theorises super-linear information-demand growth and identifies an "accuracy cliff" at capacity overflow: "when the task's information demand surpasses the model's output capacity, performance does not degrade gracefully but instead collapses sharply." The behavioural prediction is a *threshold*, not smooth $(1-\varepsilon)^n$. A cliff at a specific capacity threshold is exactly the manifold-transition behaviour our framework predicts at the boundary between covered and uncovered modes; Wan et al.'s mechanism (capacity overflow) is one specific cause, complementary to ours. *Where the framework concedes*: Wan documents a mechanism by which $|C|$ effectively exceeds the model's coverage in a single forward pass; Pattern A interventions (constrained decoding, formal verification) are the kind of response the framework expects but does not automatically supply.

Karpinska et al. [2024]. GPT-4o achieves 55.8% pair accuracy across 1,001 minimally-different true/false claim pairs about long fictional books (mean length 127K tokens). The paper reports performance by *evidence scope*, not by context length: 59.8% on sentence-level retrieval, 47.6% on passage-level, 41.6% on global reasoning. No per-context-length curve is reported. The decay axis is the number of evidence pieces that must be integrated—a k-axis quantity, not an n-axis one. NoCha is therefore not counter-evidence to a sublinear-in-n claim. Evidence-scope decay is exactly the k-axis observation the framework predicts cannot be addressed by scaling raw context length—it requires retrieval or process supervision along the actual decay axis. The relocation is informational, not magical.

Unifying observation. Every counter-paper decays over a variable that is not n: compositional graph size, fact count, log-time horizon, capacity threshold, or evidence scope. The apparent rapid decay is, in each case, in a quantity our framework already concentrates the action in (k_{hard} and $|C|$), not in raw sequence length. This is a relocation, not a dissolution. Where k_{hard} grows with task length—adversarial compositional structure, multi-hop fact chains, long horizons that force more decisions—reliability remains hard; the framework's value is directing intervention toward capability provisioning along the actual decay axis rather than toward context-window or compute-budget expansion that does not help.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and §1 state five contributions: (i) the three-layer framework partitioning key tokens into easy/hard subsets with a bounded mode catalogue (§3); (ii) the empirical postulate of logarithmic mode discovery anchored against three taxonomies with $\sigma \in [0.87, 1.85]$ (§3.2); (iii) Proposition 1 on the polylog intervention budget, with explicit per-hard-token vs. sequence-level distinction; (iv) evidence synthesis for clustering, cluster-selective interventions, and sublinear length scaling (§4); (v) the counter-evidence re-audit (Appendix C). Each claim is developed in the cited section. The paper makes no original empirical claims and is scoped as a framework-plus-synthesis contribution; limitations are explicit (§6).

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: §6 enumerates eight specific limitations: (i) Postulate 1 is empirical, not derived—a genuinely Heaps-power-law domain would invalidate the doubly-logarithmic special case (while preserving the qualitative polylog conclusion of Appendix A); (ii) the coverage form is an empirical best-fit, not theoretically derived; (iii) cluster-orthogonality is approximate, with Le [2026]'s prompt-channel-interference caveat noted; (iv) domain narrowness—the empirical anchor rests on three taxonomies covering general/code/math, with agentic and scientific regimes untested; (v) the $\sigma \approx 1.85$ estimate is single-point calibration, not a discovery-curve fit; (vi) β is latent and we report no empirical range; (vii) the L2–L3 taxonomy-granularity gap is unmeasured; (viii) patch-shift between deployments changes $A_D, \sigma_D, \beta_D, |C_D|$ simultaneously. The closing "goalpost-relocation caveat" makes explicit that the framework *relocates* the difficulty rather than resolving it: where k_{hard} grows with task length, reliability remains hard, and the framework's value is in naming the on-axis intervention.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.

- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper develops a descriptive/conditional framework rather than formal theorems. Proposition 1 is engineering math whose conditional nature is stated explicitly in the surrounding "Conditionality" paragraph and in its statement: it holds conditional on (i) the log-coverage form of §3.3 on its declared domain $m \geq 2$, $|C_{\text{eff}}| \geq 2$; (ii) the residual target $\varepsilon \in (0, e_{\text{hard}})$; (iii) Assumption 2 (per-sequence active-mode exposure) for any sequence-length rate claim. The two-line proof immediately following the proposition statement traces all algebraic steps. Postulate 1 is explicitly labelled an empirical postulate, not a theorem (this is the load-bearing math-claims discipline of the paper). We label the budget statement a "Proposition" rather than a "Theorem" to keep its conditional nature visible.

   Guidelines:

   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: The paper is a framework-plus-synthesis contribution, not an empirical paper; it runs no new experiments and trains no new models. Every numerical value cited (§4, Appendix B) is pinned to a specific published source with the relevant benchmark, model, and result identified. The σ calibration in §3.2 cites ErrorAtlas, HumanEval, and MWPES with the endpoint corpus sizes and category counts used. The 28 capability-elimination citations in Appendix B report pre/post percentages from the original sources. A reader following each citation can verify or reproduce the underlying empirical result by consulting the cited work.

   Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [N/A]

   Justification: The paper releases no new datasets, models, or code. It is a framework-and-synthesis paper; all underlying empirical results are owned by their respective authors and accessed through the cited references.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.

- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: §4 specifies which benchmarks supply evidence for each load-bearing claim: ErrorAtlas (83 models $\times$ 35 datasets) and MWPES (304,865 errors, 15 models) for clustering; the 28 capability-elimination interventions (Appendix B) for cluster-selectivity, with per-row pre/post percentages and the model/benchmark named; Loong, GSM-∞, RULER, Anchor LLMs, and Press et al. for sublinear length scaling, each with the specific length range, model, and metric identified. The σ calibration in §3.2 cites three taxonomies with k_{hard} and $|C|$ endpoint values used in the formula. No original training or hyperparameter selection is performed by this paper.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [N/A]

Justification: The paper reports no original experimental results, hence no error bars. Where numerical results are cited from the literature (e.g., the $+41.4$-pp PAL gain, the 100% Acurai elimination on RAGTruth with 95% CI 91–100%, the 32.9% Loong endpoint), the originating work is cited with its reported significance treatment.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [N/A]

Justification: The paper performs no original computational experiments. Compute requirements of the cited systems are documented in those works.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: [Yes]

Justification: We have reviewed the NeurIPS Code of Ethics. The work is theoretical/synthesis with no human subjects, no scraped or sensitive data, no model training, and no deployed system. There are no fairness, privacy, or security risks specific to the paper.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: §5 discusses positive implications: targeted intervention-library design, SLA alignment between per-hard-token and sequence-level reliability targets, and parallelisable library construction. The negative-impact direction is structurally limited because the paper releases no model or dataset; the closing "goalpost-relocation caveat" (§6) explicitly warns that the framework *relocates* the difficulty of long-context reliability and should not be misread as a license to deploy LLMs in higher-stakes settings than the cited evidence justifies.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate

to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.

- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: No new pretrained models, generative artifacts, or scraped datasets are released. The contribution is a theoretical framework and a synthesis of cited work.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [N/A]

Justification: The paper uses no datasets, code, or models—it cites published work but does not redistribute any artifact requiring license disclosure. Citations follow NeurIPS conventions.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.

- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [N/A]

    Justification: No new assets are released by this paper.

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: The paper does not involve crowdsourcing or research with human subjects.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: No human subjects research is conducted, so no IRB approval is required.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: LLMs were used only for writing assistance (drafting, polishing, copy-editing) and not as part of the research methodology. The framework, the synthesis, and the choice of cited evidence are the authors'; LLMs were not involved in selecting evidence, deriving the framework, or shaping the argument. Per the question's stated guidance, no declaration is required for editing-only usage; we answer [N/A] accordingly.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.